

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Pharm. Examination

University Backlog Theory Examination January- 2018

PH114 & PH123 Pharmaceutical Chemistry-I[New Syllabus]

Date: 03/01/2018, Wednesday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

Q. 1 Attempt all questions. Each question is of one mark.

20

1. Which of the following is not used in limit test of chloride?
[A] Dilute HCl
[B] Dilute HNO₃
[C] AgNO₃
[D] None of the above
2. Hypokalemia means
[A] Higher serum potassium level than that of normal
[B] Same serum potassium level than that of normal
[C] Lower serum potassium level than that of normal
[D] None
3. Sodium chloride is prepared from
[A] Sodium carbonate
[B] Sodium Phosphate
[C] Magnesium carbonate
[D] None of the above
4. Which of the following is not used as electrolyte replenisher?
[A] Sodium chloride
[B] Talc
[C] Potassium chloride
[D] Magnesium sulphate
5. Impurities in pharmaceutical substances are generally expressed as?
[A] ppm
[B] ppb
[C] ppt
[D] None of the above

6. Gutzeit Test is involved in limit test of?
[A] Iron
[B] Arsenic
[C] Chloride
[D] Lead
7. Which of the following are units are used for expressing normal concentration of electrolytes in body fluids?
[A] mg/L
[B] gm/L
[C] kg/L
[D] mEq/L
8. Calcium gluconate is assay by which method?
[A] Complexometric titration
[B] Nitrite titration
[C] Mohr's method
[D] None of the above
9. Green vitriol is
[A] Calcium oxide
[B] Ferrous sulphate
[C] Ferric oxide
[D] None of the above
10. Which of the followings is not component of Barium sulphate reagent?
[A] Barium chloride solution
[B] AgNO_3 solution
[C] Potassium sulphate
[D] Sulphate free alcohol
11. One of the following is well in cyanide poisoning
[A] Sodium nitrate
[B] Sodium Thiosulfate
[C] Sodium nitrate and Sodium Thiosulfate Both
[D] None of above
12. The unit of viscosity is _____.
[A] Mass X Length X Time
[B] $\text{Mass X Length X Time}^{-1}$
[C] $\text{Mass X Length}^{-1} \text{ X Time}^{-1}$
[D] $\text{Mass X Length}^{-1} \text{ X Time}$
13. Radiopharmaceuticals use in diagnosis of pernicioal anaemia
[A] Ca-45
[B] Cr-51
[C] Co-57
[D] K-42

14. The force per unit length over the surface of liquid is known as:
[A] Viscosity
[B] Surface tension
[C] Osmotic pressure
[D] Gravitational pull
15. "It is only the absorbed light radiations that are effective in producing a chemical reaction."
This is the statement of
[A] Lambert law
[B] Lambert-Beer law
[C] Grothus-Draper law
[D] Stark-Einstein law
16. Graham's law refers to:
[A] Boiling point of water
[B] Volume changes of gases
[C] Gas compression
[D] Gaseous diffusion
17. In photochemical reactions, the absorption of light takes place in
[A] primary processes only
[B] secondary processes only
[C] either primary or secondary process
[D] both primary and secondary processes
18. The boiling point of a liquid is that temperature at which
[A] the vapour pressure of the liquid is equal to the atmospheric pressure
[B] the vapour pressure of the liquid is less than the atmospheric pressure
[C] the vapour pressure of the liquid is greater than the atmospheric pressure
[D] the vapour pressure of the liquid is equal to the square root of the atmospheric pressure
19. Which is the right formula for a ring detachment method for measurement of surface tension?
[A] $\gamma = \frac{F}{4\pi R}$
[B] $\gamma = \frac{F}{2\pi R}$
[C] $\gamma = F \cdot 2\pi r$
[D] $\gamma = F \cdot 4\pi R$
20. The substances which initiate a photochemical reaction but itself does not undergo any chemical change is called
[A] catalysis
[B] fluorescent
[C] sensitizer
[D] none of these

SECTION – II

Q 2 Attempt any TWO of the following:

- A Describe limit test of iron. 05
- B Enlist sources of impurities in pharmaceutical substance. Discuss any one. 05
- C Outline the method of preparation, uses and assay of Copper Sulphate. 05

Q 3 Attempt any TWO of the following:

- A Write a detailed note on ORS. 05
- B Describe biological importance of phosphate in our body. 05
- C State the Pharmaceutical Applications of radioactive substances. 05

SECTION – III

Q 4 Attempt any FOUR of the following:

- A Define the following terms: 05
(1) Astringent, (2) Haematinics, (3) Expectorant, (4) Emetic and 5) Cathartics
- B Explain Jablonski diagram of photochemistry. 05
- C Write a note on Joule-Thomson effect. 05
- D Describe physiological importance of calcium in detail. 05
- E Describe preparation, assay and storage condition for Nitrogen as important gas in pharmacy. 05
- F Enlist Colligative properties. Explain elevation of Boiling point. 05

Q 5 Attempt any FOUR of the following:

- A Describe drop weight method and drop number method for determination of Surface Tension. 05
 - B Explain: Importance of Chloride as an electrolyte. 05
 - C Differentiate Ideal Solution and Real solution in detail. 05
 - D What is viscosity co-efficient? Describe Ostwald's viscometer for determining it. 05
 - E Debye Huckel theory. 05
 - F State and explain first law of thermodynamics. 05
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